

# Design Report (Final) for EdX Phot1x

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**Abstract** This report contains a brief introduction to silicon photonics theory and application [1], followed by a description of the layout to be fabricated in the course. MODE field and circuit simulation results of the interferometer circuits in the layout, along with analysis with measured data from the fabricated device are reported.

## 1 Theory

The compact model for the effective index of an Si waveguide can be expressed by a 2nd-order polynomial. The equation is given below.

$$n_{eff} = n_1 + n_2 * (\lambda - \lambda_0) + n_3 * (\lambda - \lambda_0)^2 \quad (1)$$

where  $n_{1-3}$  are coefficients extracted by curve-fitting simulated or measured values of  $n_{eff}$  as a function of  $\lambda$  and  $\lambda_0$  is the design frequency.

In this project we will design photonic circuits based on Mach-Zehnder interferometer (MZI). Specifically we will use imbalanced MZIs where the length of the two arms are different, and the length difference  $\Delta L = L_2 - L_1$ . If the waveguide mode propagation constant is  $\beta = \frac{2\pi n}{\lambda}$ , the output intensity of an imbalanced MZI is given by

$$I_0 = \frac{I_i}{2} [1 + \cos(\beta \Delta L)] \quad (2)$$

This gives a sinusoidal output as a function of wavelength with peaks and constructive interference and zeros at destructive interference. The period of the sinusoidal output i.e the distance between two peaks is called the free spectral range (FSR) of an MZI, which is a parameter controlled by the arm length difference

( $\Delta L$ ), that shifts the phase of the wave propagating through the longer waveguide. The FSR ( $\Delta\lambda$ ) of an MZI can be expressed by the equation below as a function of  $\Delta L$ , group index  $n_g$ , and operating wavelength  $\lambda$ .

$$\Delta\lambda = \frac{\lambda^2}{\Delta L n_g} \quad (3)$$

The group index,  $n_g$  is defined by  $n_g = n - \lambda \frac{dn}{d\lambda}$

## 2 Modeling and Simulation

For this design project Silicon waveguides with SiO<sub>2</sub> cladding layers are used. Default waveguide parameters are width 500nm and height 220nm. A simulation of the TE and TM mode profiles of the waveguide was performed in Lumerical MODE solver and the results are shown below.

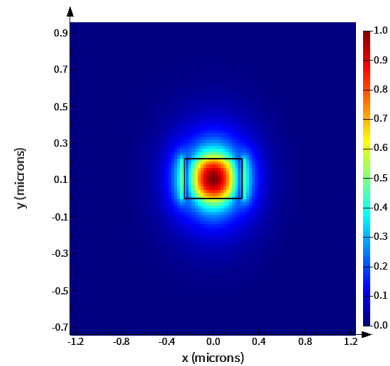


Fig. 1: Simulated TE mode field profile of Si waveguide.

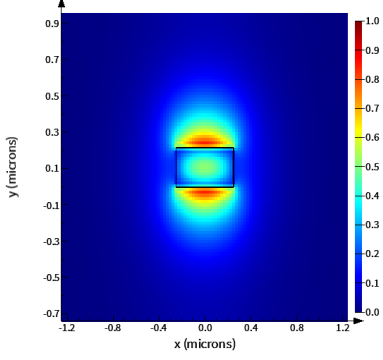


Fig. 2: Simulated TM mode field profile of Si waveguide.

The effective index,  $n_{eff}$ , and group index,  $n_g$  of the waveguide (TE0 mode) as a function of wavelength in the range of 1.5-1.6  $\mu\text{m}$  are extracted using a frequency sweep and the plots are shown below. At the design

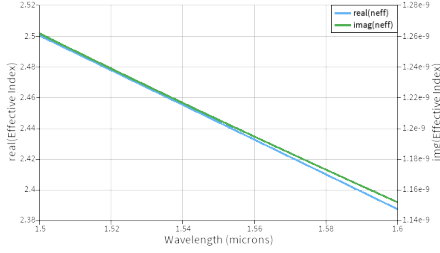


Fig. 3: Effective index of the fundamental TE mode as a function of wavelength.  $n_{eff}$  decreases with increasing wavelength.

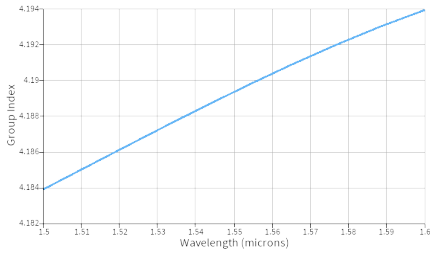


Fig. 4: Group index of the fundamental TE mode as a function of wavelength.  $n_g$  increases with increasing wavelength.

wavelength of 1.55  $\mu\text{m}$ , the TE0 mode has  $n_{eff}=2.44389$

and  $n_g=4.1894$ . From the frequency sweep, coefficients of the compact waveguide model can be extracted by curve-fitting. The compact model is given by Eq. 4.

$$n_{eff} = 2.44389 - 1.12604 * (\lambda - \lambda_0) - 0.0329638 * (\lambda - \lambda_0)^2 \quad (4)$$

where  $\lambda_0=1550\text{nm}$  is the design wavelength.

In the layout, we are provided with a 605  $\mu\text{m}$  by 410  $\mu\text{m}$  area on the chip to house our designed circuits. For this project, we are placing several MZI circuits along with a de-embedding (loopback) structure with only 2 fiber grating-couplers connected by a waveguide to measure the insertion loss. A screenshot of the layout GDS file is shown below. We have used strip Si

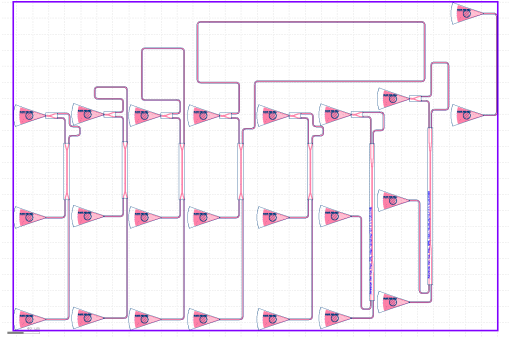


Fig. 5: GDS layout of MZI circuits. There are 7 MZIs from left to right, and a de-embedding structure at the upper-right corner.

waveguide, y-splitter, bi-directional and adiabatic couplers, and grating couplers for designing the circuit with compact models from the SIEPIC PDK for TE mode and 1550 nm wavelength. The layout design is done in KLayout, and circuit simulations are performed using Lumerical Interconnect. To make the circuits measurable by the automated equipment, the distance between the fiber grating couplers are set at 127  $\mu\text{m}$ , matching the pitch of the fiber array in the equipment. The first 5 MZIs are designed using bi-directional couplers for the output splitter, with gradually increasing 2nd arm length to vary  $\Delta L$ . MZI1 and MZI5 are identical circuits to check the repeatability of the fabrication process. MZI6 and MZI7 are designed by using adiabatic couplers instead of directional coupler, with 2 different  $\Delta L$ s. Finally, the de-embedding structure is also included to measure the insertion loss and adjust the MZI transfer functions. The minimum  $\Delta L$  is kept at  $\sim 30 \mu\text{m}$  to keep the max FSR below 25nm, as the grating couple bandwidth is 50nm. A summary of the MZIs and corresponding FSR values are shown in table 1.

Circuit simulation results for MZI1, MZI4, and MZI6 are shown below

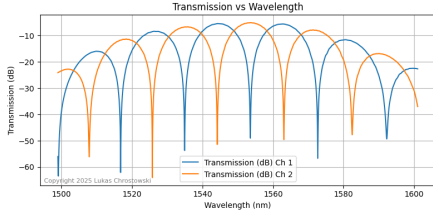


Fig. 6: Simulated transfer function of MZI1.

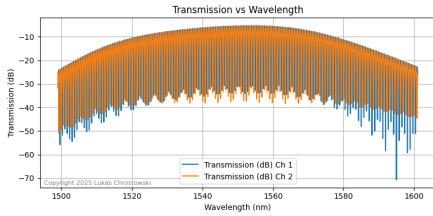


Fig. 7: Simulated transfer function of MZI4.

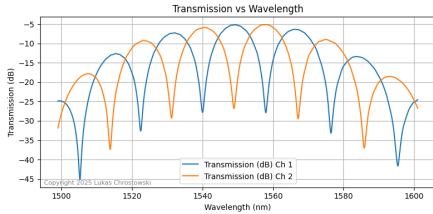


Fig. 8: Simulated transfer function of MZI6.

### 3 Fabrication

#### 3.1 Applied Nanotools, Inc. NanoSOI process:

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness, and 2  $\mu\text{m}$  buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed

into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1  $\text{H}_2\text{SO}_4:\text{H}_2\text{O}_2$ ) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a Raith EBPG 5000+ electron beam instrument using a raster step size of 5 nm. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2  $\mu\text{m}$  oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300°C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery. Some SEM images of the devices fabricated in the October 2025 run are shown in figure 9.

### 4 Experimental Data

To characterize the devices, a custom-built automated test setup [2,6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90° rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fiber array was used to couple light in/out of the chip [5].

For simplicity, we will plot measurement data for the circuits MZI3 ( $\Delta L = 262.2\mu\text{m}$ ) and MZI7 ( $\Delta L = 129.4\mu\text{m}$ ). Figure 10 shows MZI3 measured vs simulated transmission. Here we add an offset to the simulated spectra to for a better comparison of the peaks. We see that the peaks line up very well, but the insertion loss profile is different between the measured and simulated spectra, showing a difference in the peak of

| Device | L1( $\mu\text{m}$ ) | L2( $\mu\text{m}$ ) | $\Delta\text{L}(\mu\text{m})$ | FSR (analytical,nm) | FSR (simulated,nm) |
|--------|---------------------|---------------------|-------------------------------|---------------------|--------------------|
| MZI1   | 39.228              | 70.862              | 31.634                        | 18.06               | $\sim 18$          |
| MZI2   | 39.228              | 177.13              | 137.902                       | 4.1                 | $\sim 4.1$         |
| MZI3   | 39.228              | 301.43              | 262.202                       | 2.2                 | $\sim 2.19$        |
| MZI4   | 39.228              | 824.223             | 784.995                       | 0.73                | $\sim 0.73$        |
| MZI5   | 39.228              | 49.428              | 10.2                          | 18.06               | $\sim 17.9$        |
| MZI6   | 41.924              | 73.558              | 31.634                        | 18.06               | $\sim 17.3$        |
| MZI7   | 41.924              | 171.366             | 129.442                       | 4.4                 | $\sim 4.4$         |

Table 1: MZI circuits and FSR from both analytical eq. 3 and calculated from Lumerical Interconnect simulation



Fig. 9: SEM images from the October 2025 ANT run.

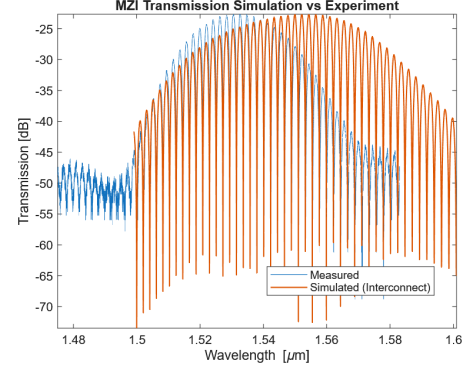


Fig. 10: Measured vs Simulated transfer function for MZI3.

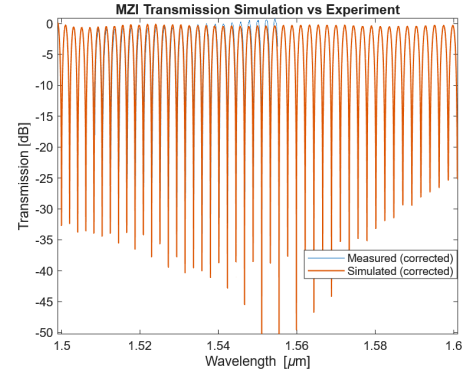


Fig. 11: Insertion loss corrected Measured vs Simulated transfer function for MZI3.

the envelope. So we correct the spectra for both cases by calibrating with the deembedding (loopback) structure transmission and replot the corrected transmission spectra in figure 11 which show a very good match. We create the same plot for MZI7 in figure 12. For MZI7 we see slight shift for the measured spectra. In the design, MZI1 and MZI5 are identical structures to check for design repeatability. In figure 13 we plot their mea-

sured transmission. The slight discrepancy between the two measurements can be attributed to the  $\sim 300\mu\text{m}$  distance between the two circuits, as there can be measurement variability across the chip area.

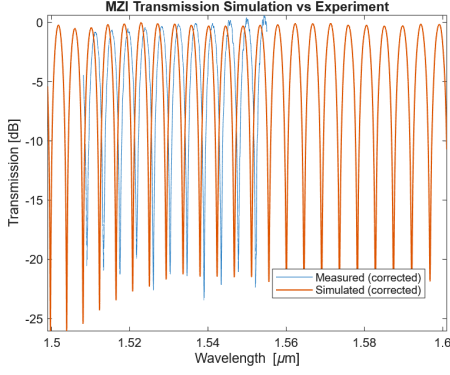


Fig. 12: Insertion loss corrected Measured vs Simulated transfer function for MZI7.

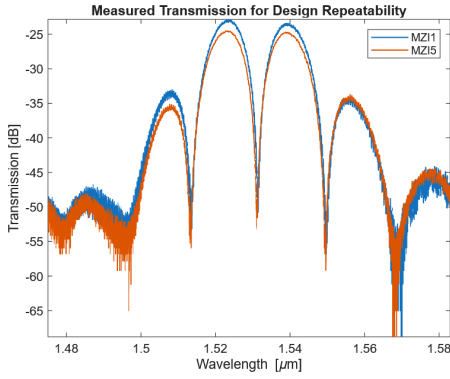


Fig. 13: Comparison of measured transmission for MZI1 and MZI5.

## 5 Analysis

### 5.1 Extracting $n_g$ and FSR from measured spectra

For measurement data analysis we follow the steps described in the **Measurement Data Analysis** section of the course. The steps are briefly as follows:

- Baseline correction: Remove the envelope due to grating couplers by curve-fitting the transmission of the loopback structure (pair of grating couplers) with a low-order ( $\leq 5$ ) polynomial and subtracting this from the device transmission to *flatten* the spectrum.
- Fit the flattened curve to the target MZI transfer function found from theoretical analysis, shown in eq. 5.

$$F = 10 \log_{10}$$

$$\left( \frac{1}{4} \left| 1 + \exp \left[ -i \frac{2\pi n_{\text{eff}}}{\lambda} \Delta L - \alpha \Delta L / 2 \right] \right|^2 \right) + b \quad (5)$$

where  $\alpha$  is the waveguide loss,  $\Delta L$  is the MZI waveguide length mismatch (constant from layout),  $b$  is an offset parameter to account for excess insertion loss [dB], and  $n_{\text{eff}}$  is given by eq. 1.

- Find initial curve-fitting parameters by calculating FSR from the measured spectra and finding  $n_g$  and  $n_{\text{eff}}$  from the FSR value. Then perform curve-fitting to fit the measured spectra using a least-squares curve-fitting algorithm (Matlab function *lsqcurvefit*).
- Finally, calculate  $n_g$  from the fitted parameters.

We go through these steps for MZI3 as an example. Figure 14 shows the curve-fitted MZI model. From this we

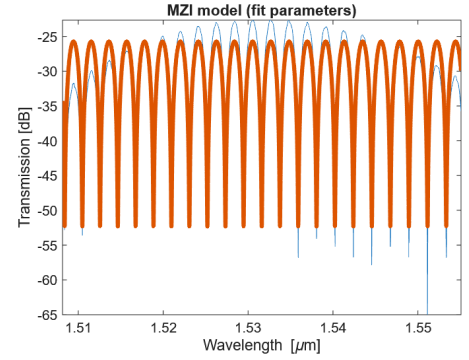


Fig. 14: MZI3 measured transmission and fitted model.

extract  $n_g$ . Figure 15 shows the extracted group index. The  $n_g$  at operating wavelength 1550nm is found to be 4.1723. We also plot the FSR extracted from simulation

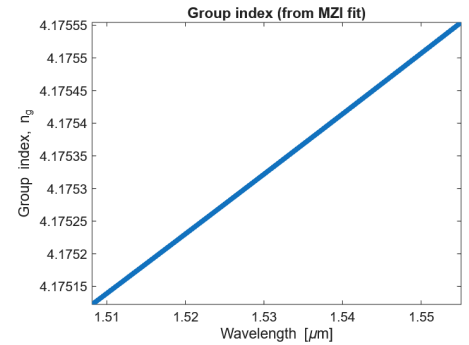


Fig. 15: MZI3 group index extracted by fitting measured data.

and measurement for MZI3 in figure 16. We repeat the calculation for all the circuits and update table 1 with measured FSR and  $n_g$  values. We plot the FSR values vs  $\Delta L$  in figure 17 to show the variation in MZI characteristics with design variable.

| Device | L1( $\mu\text{m}$ ) | L2( $\mu\text{m}$ ) | $\Delta L(\mu\text{m})$ | FSR (analytical, nm) | FSR (simulated, nm) | FSR (measured, nm) | $n_g$ (measured) |
|--------|---------------------|---------------------|-------------------------|----------------------|---------------------|--------------------|------------------|
| MZI1   | 39.228              | 70.862              | 31.634                  | 18.13                | $\sim 18$           | $\sim 18.06$       | $\sim 4.067$     |
| MZI2   | 39.228              | 177.13              | 137.902                 | 4.16                 | $\sim 4.1$          | $\sim 4.08$        | $\sim 4.164$     |
| MZI3   | 39.228              | 301.43              | 262.202                 | 2.19                 | $\sim 2.19$         | $\sim 2.14$        | $\sim 4.175$     |
| MZI4   | 39.228              | 824.223             | 784.995                 | 0.73                 | $\sim 0.73$         | $\sim 0.714$       | $\sim 4.184$     |
| MZI5   | 39.228              | 49.428              | 31.634                  | 18.13                | $\sim 17.9$         | $\sim 18.096$      | $\sim 4.067$     |
| MZI6   | 41.924              | 73.558              | 31.634                  | 18.13                | $\sim 17.3$         | $\sim 16.412$      | $\sim 4.74$      |
| MZI7   | 41.924              | 171.366             | 129.442                 | 4.43                 | $\sim 4.4$          | $\sim 4.306$       | $\sim 4.197$     |

Table 2: MZI circuits and FSR from analytical eq. 3, calculated from Lumerical Interconnect simulation, and extracted FSR and  $n_g$  values from measured data.

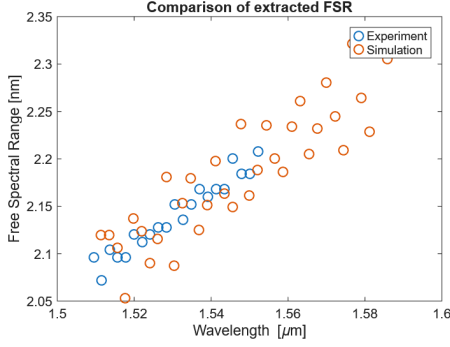


Fig. 16: MZI3 FSR comparison measured vs simulated.

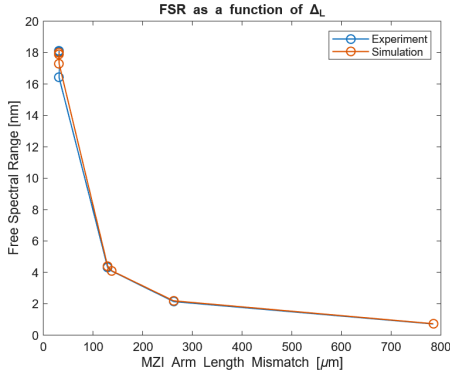


Fig. 17: MZI3 FSR comparison measured vs simulated.

## 5.2 Corner Analysis

To account for fabrication variability, we perform corner analysis on the waveguide width and height parameters. We consider maximum and minimum width of 520nm and 480nm, with nominal width 500nm. For height we consider maximum and minimum of 223.1nm and 215.3nm, with nominal height of 220nm. We perform eigenmode analysis in Lumerical MODE and extract the  $n_g$  for each design. Table 3 shows the designs considered for corner analysis and corresponding  $n_g$  values at  $\lambda_0=1550\text{nm}$ . From the corner analysis the

maximum and minimum  $n_g$  values are 4.228975 and 4.152919, with nominal value at 4.18974. From table 2, we compare the extracted  $n_g$  values with these corner limit values (see figure 18). We see that group indices of all circuits except MZI1, MZI5, and MZI6 fall within this range. The reason could be a poor curve-fit for these circuits due to not enough peaks being available, which resulted in poor model fit for  $n_g$  extraction.

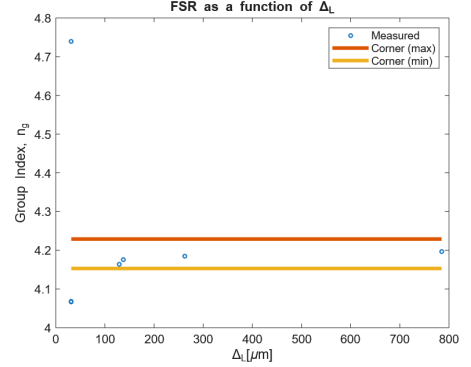


Fig. 18: Measured group index plotted vs  $\Delta L$  with the corner analysis limits.

## 6 Conclusion

In this report we designed 7 different MZI circuits using SIEPIC PDK in KLayout, with different  $\Delta L$  parameters, simulated the circuits in Lumerical Interconnect, and performed eigenmode analysis of the waveguide structures in Lumerical MODE. We analyzed the measured data and compared with simulation. We also extracted free spectral range of the MZIs from the measured data using curve-fitting, and performed corner analysis for the group index due to waveguide fabrication variability.

| Design        | Width (nm) | Height (nm) | $n_g$ (simulated) |
|---------------|------------|-------------|-------------------|
| Nominal (N)   | 500        | 220         | 4.18974           |
| Corner 1 (C1) | 520        | 223.1       | 4.159804          |
| Corner 2 (C2) | 520        | 215.3       | <b>4.152919</b>   |
| Corner 3 (C3) | 480        | 215.3       | 4.221014          |
| Corner 4 (C4) | 480        | 223.1       | <b>4.228975</b>   |

Table 3: Corner Analysis to find limits of  $n_g$  variation with fabrication variability.

## 7 Acknowledgements

We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Cameron Horvath at Applied Nanotools, Inc. Richard Mateo Branion-Calles performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Python, and KLayout for the design software.

## References

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